

Definition

A **linear function** has the form

$$f(x) = mx + b,$$

where m is the slope and $(0, b)$ is the y -intercept. Its graph is a straight line and its domain is $(-\infty, \infty)$.

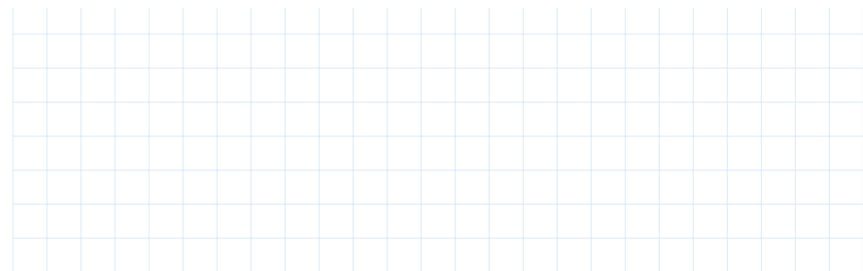
Key features

- constant rate of change m ;
- $m > 0$: increasing;
- $m < 0$: decreasing;
- $m = 0$: constant;
- zeros occur where $f(x) = 0$.

First example

$$f(x) = -3x + 6$$

Identify the slope, y -intercept, direction, and zero.



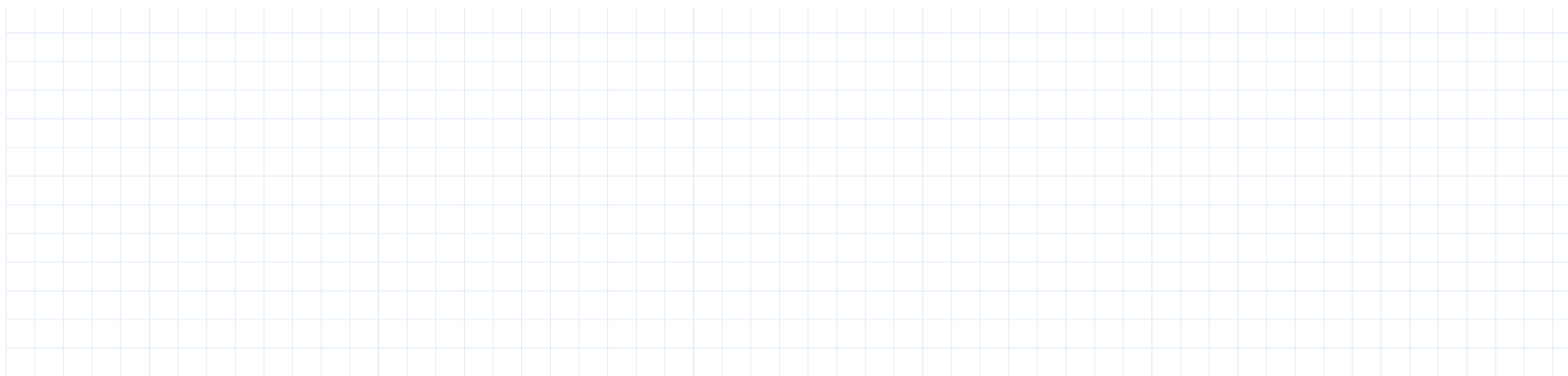
Slope and Rate of Change

For two distinct points (x_1, y_1) and (x_2, y_2) ,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} \quad (x_1 \neq x_2).$$

For a linear function, this value is the same between *every* pair of points.

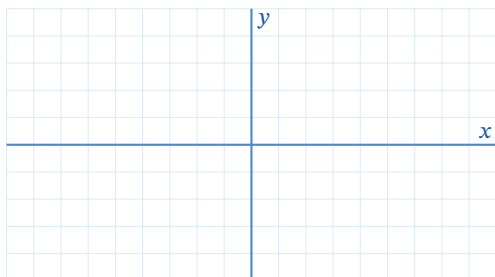
Example. Find the slope of the line through $(-2, 5)$ and $(4, -1)$. Then interpret its sign.



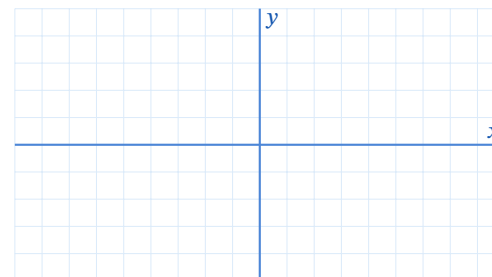
Read the Sign of the Slope

For each graph, decide whether the slope is positive, negative, or zero. Then sketch one possible line.

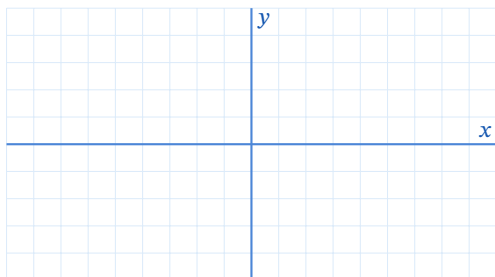
1. Increasing



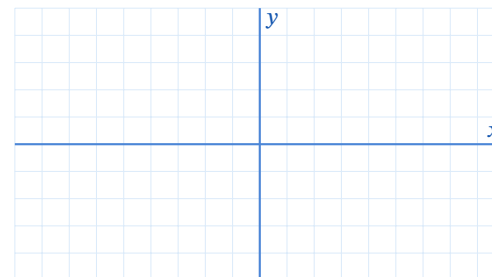
2. Decreasing



3. Constant



4. Vertical line: is it a function?



Linear or Nonlinear from a Table

Compute the change in y for each equal change in x . A constant rate of change signals a linear function.

Table A

x	-1	0	1	2
$f(x)$	0	4	8	12

Linear or nonlinear? If linear, find m .

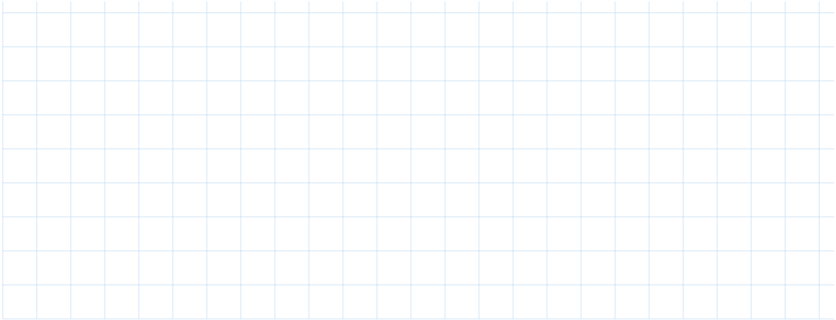
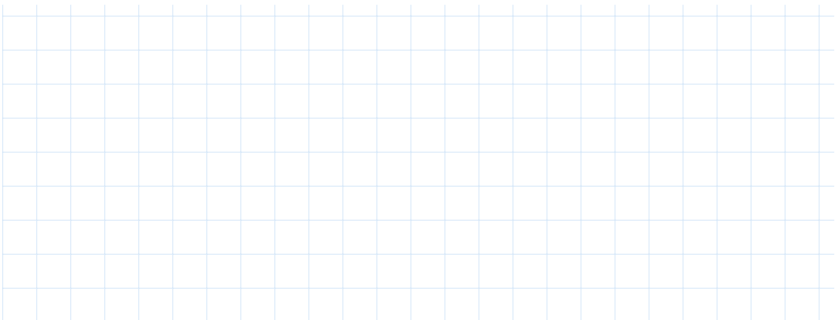


Table B

x	-2	-1	0	1	2
$g(x)$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4

Linear or nonlinear? Explain using rates of change.



Write a Linear Equation

Choose a form that matches the information given.

Information	Useful form	What to do
Slope and y -intercept	$y = mx + b$	Substitute m and b .
Slope and one point	$y - y_1 = m(x - x_1)$	Use point–slope form, then simplify if needed.
Two points	$m = \frac{y_2 - y_1}{x_2 - x_1}$	Find m first, then use either point.

Example. Find the equation of the line through $(-1, 2)$ and $(3, 10)$.



Zeros and Intersections

Zero of a linear function

To find the zero of f , solve

$$f(x) = 0.$$

The corresponding graph point is an x -intercept $(x, 0)$.

Example: $f(x) = 4x - 1$.

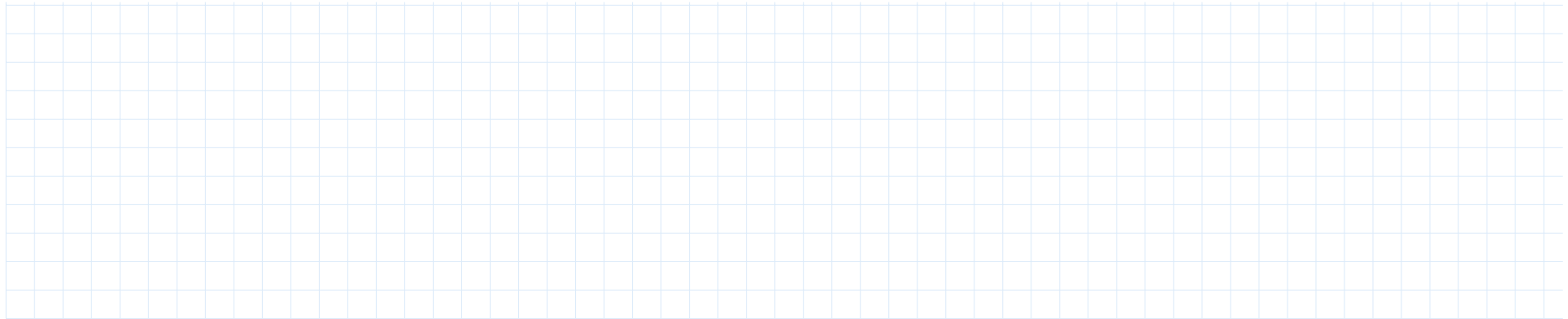
Intersection of two functions

To find where f and g meet, solve

$$f(x) = g(x),$$

then substitute the x -value to find y .

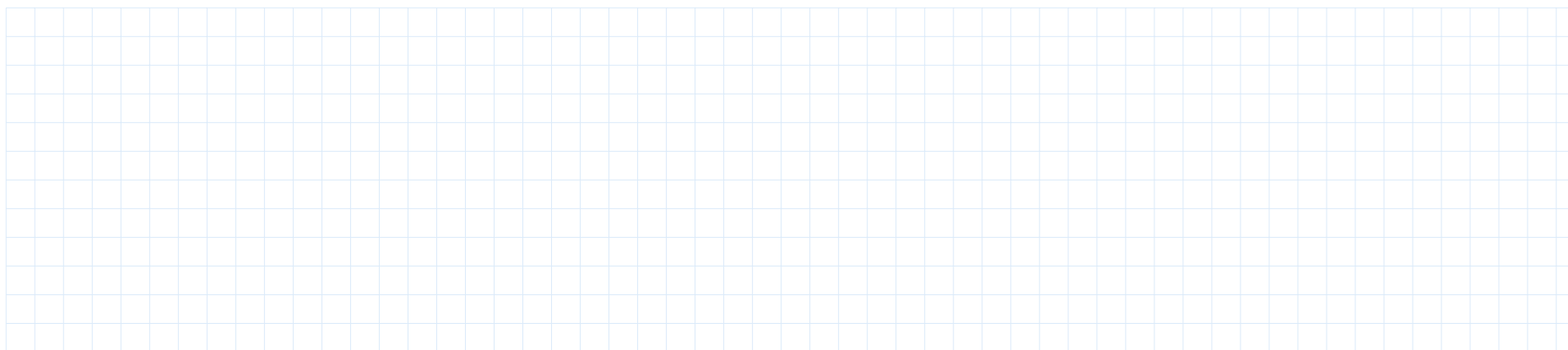
Example: $f(x) = 4x - 1$ and $g(x) = -2x + 5$.



Solve an inequality using the same algebra as an equation, with one crucial rule:

Multiplying or dividing by a negative number reverses the inequality sign.

1. If $f(x) = 4x - 1$, solve $f(x) > 0$.
2. If $f(x) = 4x - 1$ and $g(x) = -2x + 5$, solve $f(x) \leq g(x)$.



Linear Model Practice

A gym charges a 25 registration fee and 18 per month.

- (a) Write the total cost $C(t)$ after t months.
- (b) Identify and interpret the slope and y -intercept.
- (c) Find $C(8)$ and explain its meaning.
- (d) A second gym charges $C_2(t) = 22t + 5$. When do the costs match?
- (e) Which plan is less expensive after one year?

